

```
! pip install tf_keras
```

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Requirement already satisfied: tf_keras in /usr/local/lib/python3.12/dist-packages (2.20.0)
Requirement already satisfied: tensorflow<2.21,>=2.20 in /usr/local/lib/python3.12/dist-packages (from tf_keras) (2.20.0)
Requirement already satisfied: absl-py>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf_keras) (1.0.0)
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Requirement already satisfied: flatbuffers>=24.3.25 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf_keras) (24.3.25)
Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf_keras) (0.5.2)
Requirement already satisfied: google_pasta>=0.1.1 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf_keras) (0.1.1)
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Requirement already satisfied: protobuf>=5.28.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf_keras) (5.28.0)
Requirement already satisfied: requests<3,>=2.21.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf_keras) (2.32.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf_keras) (75.1.0)
Requirement already satisfied: six>=1.12.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf_keras) (1.16.0)
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Requirement already satisfied: wrapt>=1.11.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf_keras) (1.11.0)
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Requirement already satisfied: tensorboard>=2.20.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf_keras) (2.20.0)
Requirement already satisfied: keras>=3.10.0 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf_keras) (3.10.0)
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Requirement already satisfied: ml_dtypes<1.0.0,>=0.5.1 in /usr/local/lib/python3.12/dist-packages (from tensorflow<2.21,>=2.20->tf_keras) (0.5.1)
Requirement already satisfied: wheel<1.0,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from astunparse>=1.6.0->tensorflow<2.21,>=2.20->tf_keras) (0.44.0)
Requirement already satisfied: rich in /usr/local/lib/python3.12/dist-packages (from keras>=3.10.0->tensorflow<2.21,>=2.20->tf_keras) (13.9.0)
Requirement already satisfied: namex in /usr/local/lib/python3.12/dist-packages (from keras>=3.10.0->tensorflow<2.21,>=2.20->tf_keras) (0.0.8)
Requirement already satisfied: optree in /usr/local/lib/python3.12/dist-packages (from keras>=3.10.0->tensorflow<2.21,>=2.20->tf_keras) (0.12.1)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.21.0->tensorflow<2.21,>=2.20->tf_keras) (3.4.0)
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Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.21.0->tensorflow<2.21,>=2.20->tf_keras) (2025.1.31)
Requirement already satisfied: markdown>=2.6.8 in /usr/local/lib/python3.12/dist-packages (from tensorboard>=2.20.0->tensorflow<2.21,>=2.20->tf_keras) (3.7.0)
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Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from tensorboard>=2.20.0->tensorflow<2.21,>=2.20->tf_keras) (0.7.0)
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Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich->keras>=3.10.0->tensorflow<2.21,>=2.20->tf_keras) (3.0.0)
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```

```
from sentence_transformers import SentenceTransformer,util
from sentence_transformers.util import paraphrase_mining
model = SentenceTransformer('all-minilm-L12-v2')
```

```
modules.json: 100% 349/349 [00:00<00:00, 41.6kB/s]
config_sentence_transformers.json: 100% 116/116 [00:00<00:00, 12.1kB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to
README.md: 100% 10.5k/10.5k [00:00<00:00, 787kB/s]
sentence_bert_config.json: 100% 53.0/53.0 [00:00<00:00, 6.41kB/s]
config.json: 100% 615/615 [00:00<00:00, 62.0kB/s]
model.safetensors: 100% 133M/133M [00:01<00:00, 1.19GB/s]
Loading weights: 100% 199/199 [00:00<00:00, 5677.85it/s]
tokenizer_config.json: 100% 352/352 [00:00<00:00, 26.0kB/s]
vocab.txt: 100% 232k/232k [00:00<00:00, 8.86MB/s]
tokenizer.json: 100% 466k/466k [00:00<00:00, 23.6MB/s]
special_tokens_map.json: 100% 112/112 [00:00<00:00, 11.7kB/s]
config.json: 100% 190/190 [00:00<00:00, 23.2kB/s]
```

```
s1="The cat is on the mat",
s2="A cat is sitting on the mat",
emb1=model.encode(s1)
emb2=model.encode(s2)
sim_score =util.cos_sim(emb1,emb2)
print(sim_score)
```

```
tensor([[0.9250]])
```

```
sentences = [
    "The cat sits outside",
    "A man is playing guitar",
```



[Show hidden output](#)

```
model = SentenceTransformer("distiluse-base-multilingual-cased-v1")
```

[Show hidden output](#)

```
corpus1 =[
    "I love playing football.",
    "I enjoy playing soccer.",
    "She is reading a novel.",
    "She is reading a book.",
    "The weather is sunny today."
]
corpus_emb = model.encode(corpus1)
```

```
cosine_similarity = util.pytorch_cos_sim(corpus_emb, corpus_emb)
avg_sim = cosine_similarity.mean(dim = 1)
outliner_index = avg_sim.argmin().item()
outliner_index
print(f"outliner sentence : {corpus1[outliner_index]} (AVG Similarity : {avg_sim[outliner_index]:.4f})")
```

```
outliner sentence : The weather is sunny today. (AVG Similarity : 0.2940)
```

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